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Force transmission through the wrist during performance of push-ups on a hyperextended and a neutral wrist

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ABSTRACT

Study Design: Cross-sectional cohort.
Introduction: Push-ups are used ubiquitously to evaluate and strengthen the upper body. They are usually performed in 1 of 2 main ways: with the wrist in hyperextension and with the wrist in a neutral position.
Purpose of the Study: The purpose of our study was to compare the dynamic forces in the wrist during the 2 push-up styles.
Methods: Fourteen volunteers performed push-ups in 2 different patterns: on a hyperextended wrist and a neutral wrist (NW). Two force plates and a motion capture system were used to measure the ground reaction forces (GRFs) and the kinematics of the upper extremity during push-ups. Kinematic and kinetic analyses were performed using Matlab software (Mathworks, Natick, MA).
Results: The GRF vector was distributed differently during the different types of push-ups. For both methods, the total GRF carried by the upper dominant extremity was larger than those of the nondominant extremity. In the NW configuration, the GRF vector was more uniform throughout the push-up in the vertical direction. The horizontal distance between the capitate bone location and the GRF origin was smaller in hyperextension. The forces traveled more dorsally over a wider area and more ulnarly in the hyperextended wrist.
Discussion: Forces are transmitted differently through the wrist in the 2 methods. Push-ups on an NW are likely safer because ligaments may be preferentially loaded in hyperextension. Further study may delineate the differences in the anatomic location of force transmission and the long-term clinical effect on the wrist.
Conclusions: This study supports the performance of push-ups on a wrist in neutral flexion extension; both to enable patients after surgery or injury to strengthen the upper body and prevent injury and long-term wear in the wrist. The knowledge gained from this study may assist in outlining guidelines for push-up performance.
Level of Evidence: Diagnostic level 2a.
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Introduction

Push-ups are used frequently for the evaluation, strengthening, and training of the upper body.[1-4] They are used in military training and in the training of both professional and amateur athletes of all kinds and ages. Used to promote fitness in general, they are an important part of many rehabilitation protocols after surgery and injuries that cause wasting or weakness of the upper extremity.[5] Furthermore, they are used for upper body strengthening in sedentary individuals engaging in moderate exercise. Slater et al[6] used push-ups to evaluate strength of the upper extremity in professional ice skaters. Smith et al[7] measured muscular strength looking at the ability to perform push-ups and grip strength as measured by a grip meter. Martin et al[8] used push-ups both in the evaluation of muscular endurance and in the training of patients after treatment for breast cancer, whereas Colado et al[9] used push-ups as part of an exercise program for postmenopausal women. Multiple studies have looked at the use of push-ups in physical fitness protocols in children and adolescents.[10-12]

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However, despite their common use, there is insufficient evidence in the literature as to how and where the forces are transferred during a push-up. Little is known about their effect in the long term and possible injuries they may cause.[13-15] Even fewer studies provide guidelines as to how they should best be performed.[16-18] This information is especially important in nonprofessional athletes who undertake push-ups to get in shape and may injure their wrists with consequences that may affect their activities of daily living. Recently, studies have begun to address these aspects, evaluating the techniques used in the performance of push-ups and their biomechanical effect. Sandhu et al[19] assessed muscle activity using surface electromyography, comparing push-ups performed on an unstable surface (Swiss ball) with those performed on a stable surface. Another study described muscle activity levels during push-ups with different loads and stability conditions again using surface electromyography. They found that suspended push-ups are a highly effective way to stimulate abdominal muscles and that there was a differential muscle stimulus depending on the push-up conditions.[3] Koch et al[20] compared vertical ground reaction force (GRF) characteristics between the clap push-up and box drop push-ups. Kim et al[21] studied the effect of different hand positions on the shoulder girdle musculature during push-ups. These studies did not distinguish between the different wrist positions.

Chuckpaiwong and Harnroongroj[22] studied the forces applied to the palm in different hand positions and found that various hand positions loaded areas of the palm differentially. The investigators concluded that the lunate area was the most loaded area in the various positions and suggested using this information to modify rehabilitation protocols.

The patterns of load on the wrist during push-up performance may cause overload on certain areas of the upper extremity, causing pain and injury. Multiple studies have described injuries related to push-up performance such as fractures and dislocations.[23-25] In addition, bearing a heavy weight on a hyperextended wrist (HW) can cause early development of wrist pain and associated carpal tunnel syndrome.[26,27]

Although studies have demonstrated clinical relevance of push-up technique, the exact anatomic passage of forces has not been elucidated. This is likely due to the difficulty in measuring and locating these forces in vivo while the wrist is actively moving and being loaded. Several studies have examined the kinematics and loading of the wrist with and without pathology ex vivo. Short et al[28] used a cadaver model to evaluate the distribution of forces and movement of the carpal bones in the wrist. Scaphoid and lunate motions along with radiocarpal and ulnocarpal pressure patterns were continually monitored while the wrist was manipulated through flexion and extension. They reported that, before ligament sectioning, the position of the scaphoid and lunate was dependent on the position of the wrist and the direction of wrist motion. Sectioning the scapholunate interosseous ligament caused increased scaphoid flexion, scaphoid pronation, and lunate extension. Pressure on the radiocarpal and ulnocarpal joints was redistributed after ligament sectioning.

Tencer et al[29] and Viegas et al[30] applied a static force to the second and third metacarpals of cadaver upper limbs. By changing the hand position, they recorded the pressure areas using pressure-sensitive paper. Their results clearly indicate that the pressure area moves across the radial bone according to the position of the hand. In addition, in the HW, the force passes more dorsally than in flexion or in the neutral wrist (NW) positions.[29,30] This result was confirmed by Majima et al[31] using an anatomic computer-aided design model built from computed tomography scans. They loaded the computer-aided design model of the wrist in the hyperextended position by applying static forces to the trapezium and hamate bones and in the neutral position by loading 5 metacarpal bones.[31] Rainbow et al[32] evaluated in vivo kinematics during extreme wrist positions. They concluded that during extreme extension, the radial-scaphoid contact area shifts dorsally. They also found that the lunate-radius and scaphoid-radius contact areas increased 45% ± 22% and 13% ± 16%, respectively, from a neutral grip position.

Clinically, we have observed that the vast majority of patients who are treated for wrist injuries are unable to hyperextend their wrists (often beyond neutral position). The term hyperextension cannot be precisely defined and likely differs between individuals. Everyday activities require a variable range of motion in the wrist. Ryu et al[33] demonstrated that most everyday activities are performed within 60° of extension, which is about 70% of the total extension of the normal wrist. Palmer et al[34] found that most functions require only 30° of extension. Still when the wrist is loaded in extension, the position goes beyond normal everyday extension of the wrist. We defined this position as hyperextension. These patients are often excused from performing push-ups because load on a HW may be painful, even in a wrist that can reach the hyperextended position. Furthermore, many of our patients with a previous wrist injury experience difficulty in performing push-ups.

Some wrist injuries (such as scaphoid fractures and dislocations) have been described as stemming from load on an extended/HW.[23,35] However, we believe that less extreme injuries occur in repetitive less extreme activities such as sports and recreation. Some studies have noted similar observations.[36,37] These problems also apply to other joints in the extremity: bilateral shoulder dislocations, stress fracture of the hamate, and wrist pain have been described along with other injuries that have occurred after different kinds of push-up exercises.[24,25,38-40]

We suggest that push-ups on a HW preferentially load different areas of the wrist when compared with push-ups performed on an NW. Because these areas include ligamentous structures stabilizing joints in the wrist, and joint synovium contains pain receptors,[41,42] this might explain the clinical observations of pain related to hyperextension of the injured wrist and the observed difficulty in performing this style of push-ups after surgery such as open reduction and internal fixation of distal radius fractures or dorsal wrist ganglion excision where the ligaments are involved in the surgery.[31,38] If, indeed, push-ups performed on an NW are healthier for wrist stability and ligamentous integrity and load the ligaments to a lesser degree, NW push-ups could then be safely used for rehabilitation and upper body strengthening in healthy athletes. This could potentially change the way we encourage push-ups to be performed, allowing for prevention of injury, as well as better and earlier rehabilitation.

The purpose of this study was to compare in vivo GRF load vectors in an NW push-up to push-ups performed on a HW. As a first step in elucidating the transfer of forces through the wrist during the performance of push-ups, we hypothesized that the GRF will be distributed differently between the different methods.

Methods

Fourteen healthy subjects were recruited for the study. The number of subjects was based on our experience in biomechanical analysis in the gait laboratory and a review of other published studies. Ethics committee approval was obtained before study commencement. Inclusion criteria included healthy male volunteers with a normal body mass index between 18.5 and 24.9 and a height between 165 and 180 cm. The subjects brought in written documentation of good physical condition and the ability to perform multiple push-ups. Only right-handed subjects were

included (the dominant hand was defined as the one used for writing). Subjects with a history of any significant upper extremity injury, including wrist, hand, and shoulder injuries or surgery were excluded. The volunteer's height, weight, and arm length were measured before the experiment.

The subjects were asked to perform push-ups in 2 different patterns: on an NW and on a HW. In both patterns, the hands were placed shoulder-width apart, heels together, with no other part of the body touching the floor during the push-up. The HW hand position was defined in the following way: thumbs are placed exactly beneath the shoulders in the up position, and the fingers are placed apart with the forearms pronated. In the NW position, the thumbs were placed beneath the shoulders in the up position, but with neutral rotation of the wrist and forearm, thus the knuckles were placed along the body of the participant. The elbow position was not determined in a strict manner during the exercise. The subjects had 2 seconds for the downward motion and 2 seconds for the upward motion for both push-up styles. Push-up rate was controlled by a metronome. Each subject performed 2 sets of 10 push-ups for both HW and NW styles (total of 4 sets). Breaks were allowed between sets, and the sets were performed in random order to prevent any bias stemming from physical difficulty or subject fatigue. The breaks between the sets were about 5 minutes long, although the participants were free to rest longer. Because the breaks were more than 5 times longer than the exercise set, and none of the participants had reported any difficulty or discomfort performing the exercises, we do not believe that muscle fatigue was an issue during the study.

The push-ups were performed in a gait analysis laboratory. Three-dimensional motion analysis was performed using an 8-MX-13 Vicon camera system (Oxford Metrics Ltd, Oxford, UK) at a sampling frequency of 120 Hz. GRFs during push-ups were sampled by 2 three-dimensional AMTI OR6-7-1000 force plates at 960 Hz. We used standard AMTI OR6-7-1000 force plates that are calibrated by the manufacturer during installation in the laboratory. According to the manufacturer, the error in position of the center of pressure (COP) is 0.4 mm and 0.25% of applied load. All the information was collected during 3 sessions on 3 separate days. Before each test, a reset was performed to the force plate making sure that they showed 0 with no load.

To capture upper limb movement, thirteen 14-mm reflective markers were attached to each of the limbs. In addition to the standard Vicon Plug-in Gate upper body model (Guide, nd), 6 extra markers were added to track specific anatomic landmarks (Fig. 1). The following marker positions were used based on the Vicon model: the lateral acromion bone (1), upper arm (deltoid insertion) (2), proximal ulna bone (olecranon—tip of the elbow) (5), forearm (radial middle forearm) (6), distal radius-radial styloid tip (10) and ulna (ulnar styloid) (9), and the little finger metacarpal head (13). Assuming that the capitate bone is the biomechanical center of the wrist and that the center of rotation of the hand is located within the capitate bone, an additional marker was attached dorsally to the skin over the center of the capitate.[43] We estimated the error of this marker to be 3 mm based on the size of the capitate and soft tissue artifact (STA) of the skin marker. The position of the center of the capitate was determined by a trained radiologist using ultrasound (Ultranonix Medical Corporation) (position 12). To accurately record rotation and anterior translation of the limbs, 2 additional markers were placed in each segment of the upper limb: the upper arm (position 3), the distal 1/3 of the humerus (position 4), forearm (positions ulna middle 1/3 [position 7]), middle forearm (position 8), the little finger metacarpal head (position 13), and the index metacarpal head (position 11). We noted that, to get proper information, the markers were placed such that they do not form a straight line with other markers in the same hand segment to avoid singular measurements (markers dependency).[44]

Data collected included force vector components and the COP for each hand. In addition, the locations of the 26 markers and/or positions attached to the upper limbs were also recorded. All information was recorded simultaneously and continuously during the test.

Data analysis was carried out using Matlab software (Mathworks, Natick, MA). We evaluated 2- and 3-dimensional data. All analyses were performed relative to the anatomic landmarks, and the results represented after rotation to the same hand position. All magnitudes of errors were derived from the measuring error of the system. For the VICON system, it was 0.1 mm and for the force plate COP 0.4 mm. The calculation of the convex hull was made by MATLAB built-in function that has a negligible numerical error compared with the measurement errors.

Diversification of the GRF was performed by computing the angle of distribution of the GRF vectors with the vertical Z-axis (normal to the laboratory floor). We also used a Matlab built-in function taken from the field of computational geometry called convex hull. The convex hull is the intersection of all convex sets containing a set of points. Using this principle, the area of the convex hull of the set of points defined by the intersection of the GRF vector with a plane parallel to the floor lab at $Z = 1$ was calculated for both push-up styles. The area of the convex hull of the GRF initial point (COP) as detected on the force plates during

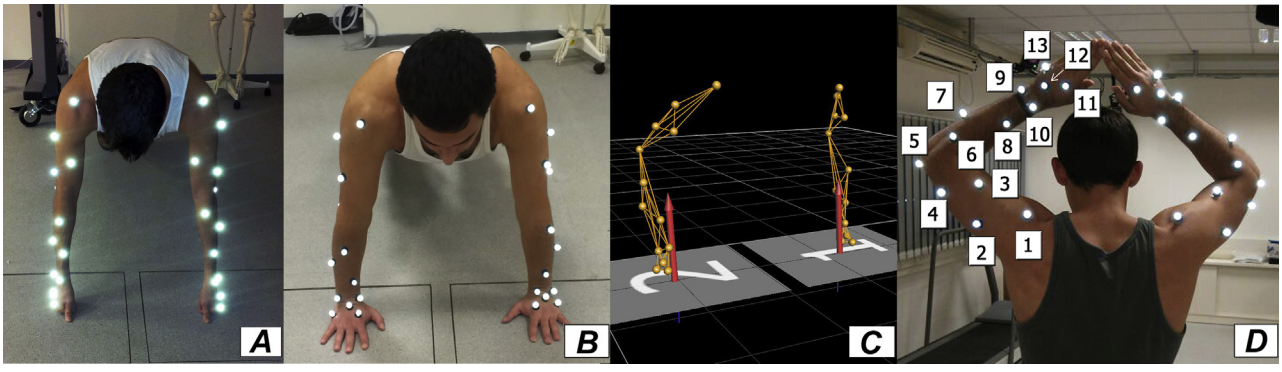


Fig. 1. (**A**) The neutral wrist (NW) push-up style. (**B**) The hyperextended wrist (HW) push-up style. (**C**) An example of a 3-dimensional reconstruction of the NW push-up kinematics (in yellow) and force measurements (red vectors) (Vicon Nexus);1 represents the left upper extremity, and 2 represents the right upper extremity on the force platforms. Note that the entire construct is shifted palmarward because push-ups done in the HW position result in a relative decrease of hand height. (**D**) Upper limb marker positions: 1—lateral acromion bone, 2-4—three markers on upper arm segment, 5—elbow bone marker, 6-8—three markers on forearm segment, 9—distal ulna bone marker, 10—distal radius bone marker, 11-13—three markers on hand segment, and in addition 12—capitate bone marker. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

push-up styles was also calculated. The 2-dimensional diversification of the capitate bone was calculated as the area of the convex hull of the capitate marker. Finally, the distance between the radius marker and the GRF COP as well as the distance between the capitate marker and the GRF COP was also calculated for both push-up styles.

Statistical analysis was performed using IBM SPSS software, version 22.0. Because each subject performed 2 sets of push-ups on a HW and 2 sets on an NW (4 sets in all), we averaged the 2 sets and then used the paired *t* test (so each subject was his own control) for variables with a normal distribution and the Wilcoxon signed rank test for nonparametric variables. These tests were used to compare values between the 2 groups (push-up styles). All statistical tests were 2-tailed, and *P* values less than .05 were considered significant.

Intrasubject reliability was assessed using the intraclass correlation coefficient (ICC) and calculated using mixed-model repeated measures.[45] In this test, the degree of reliability is expressed as a ratio of subject variability to total variability. ICC $\geq$ 0.7 is desirable and indicates good reliability.

As stated, a 2-tailed *t* test was used to evaluate the magnitude of difference between the variables in the 2 groups (NW vs HW). The direction of the difference—that is if 1 group was larger or smaller—was evaluated separately for each variable.

Results

The mean age of the subjects was 24.86 (0.95) years. Average height was 176.09 (3.06) cm, and the mean body mass index was 22.94 (1.42). All calculations of ICC showed good agreement between tests (Table 1). The total GRF measured did not differ between the 2 patterns of push-ups ($P = .07$). The upper limbs were loaded with 45%-91% of total body weight with an average of 70% during the experience. The applied GRF on the dominant hand was higher regardless of push-up style ($P < .006$).

Averaging all participants using the ulna and radius markers, in hyperextension, the wrist was pronated on average of 85.93° (standard deviation [SD] = 9.04) and 23.93° (SD = 11.07) in the NW position. The average forearm rotation differed between the 2 push-up styles by 69.69° (SD = 13.24).

All calculations were performed on both hands; because the results were similar for dominant and nondominant hands, without loss of generality, we present the results for the dominant hand.

The angles between the GRF vector and the Z-axis (normal to the force plates) differed significantly between the 2 push-up styles. That is, the range of the angles during NW push-ups was significantly smaller than those of the HW push-ups ($P < .00002$) (Table 1). This finding was supported by the area of the convex hull of the intersection of the GRF vectors with the Z = 1 plane. The area of the convex hull for the NW push-up style was significantly smaller than that of the HW push-ups ($P < .00001$) (Fig. 2). Figure 3 demonstrates the forces during a push-up cycle.

The area of the convex hull of the GRF vector's COP, as detected on the force plates (Z = 0 plane), was significantly smaller for the NW push-ups compared with the HW push-up style ($P = .0076$) (Fig. 2; Table 1). The area of the convex hull of the projection of the capitate marker on the Z = 0 plane was calculated for both NW and HW push-up styles. The convex hull area was significantly smaller for the NW push-ups than for the HW push-ups ($P = .001$) (Fig. 2). The area of the convex hull of the projection of the radius and ulna markers on the force plates was evaluated. There was no significant difference in the convex hull area between push-up styles. In addition, the area of the convex hull of the projection of the hand markers (positions 11 and 13) was evaluated. There was a significant difference in the convex hull of these markers in different push-up styles ($P < .05$). The marker movement was more significant in HW, indicating that the hand moves more during the exercise.

The COP of the GRF vector was localized with respect to the projection of the 2 bony landmarks of the radius and ulna, on the Z = 0 plane. These were found to be relatively constant throughout the push-up, regardless of push-up style.

The absolute distance between the capitate marker and the GRF initial point was significantly smaller and found to be more dorsal during HW push-ups than during NW push-ups ($P < .000001$) (Fig. 2). The absolute distance between the radius marker and the GRF COP was significantly smaller during the NW push-ups than with the HW push-ups ($P = .0001$) (Fig. 2), suggesting that the GRF passed significantly closer to the radius during NW push-ups and closer to the ulna during HW push-ups (Fig. 2).

Table 1
Differences between push-up styles

Description	Hand	Type	Mean	SD	*P*	Statistical test	ICC[a]
Total GRF compared in left hand in 2 different push-up patterns (N)	LH	NW	241.6	19.02	.03627	T	0.94
	LH	HW	246.73	16.69			0.90
Total GRF compared in 2 hands NW (N)	LH	NW	241.6	19.02	.00022	T	0.94
	RH	NW	260.33	14.67			0.94
Total GRF compared in 2 hands HW (N)	LH	HW	246.73	16.69	.00572	T	0.90
	RH	HW	258.02	14.09			0.90
GRF COP convex hull area (mm^2)	RH	NW	152.92	95.39	.00763	W	
	RH	HW	273.79	124.66			
Capitate bone marker convex hull area (mm^2)	RH	NW	42.78	41.87	.00125	T	
	RH	HW	99.12	54.63			
Index finger metacarpal head marker (position 11) convex hull area (mm^2)	LH	NW	19.55	17.56	.00993	W	
	LH	HW	43.71	25.26			
Little finger marker (position 13) convex hull area (mm^2)	LH	NW	9.4	11.08	.01310	W	
	LH	HW	27.18	19.93			
Distance between radial styloid marker and GRF COP (mm)	RH	NW	16.16	5.87	.00012	T	0.96
	RH	HW	27.77	4.25			0.75
Distance between capitate bone marker and GRF COP (mm)	RH	NW	31.99	2.11	<.00001	T	0.69
	RH	HW	15.94	3.54			0.85

SD = standard deviation; ICC = intraclass correlation coefficient; GRF = ground reaction force; LH = left hand; NW = neutral wrist; T = paired 2-tailed *t* test; HW = hyperextended wrist; RH = right hand; COP = center of pressure; W = paired 2-tailed Wilcoxon signed rank test.
Comparison between forces among all participants' data: force angle, anatomic landmarks, convex hull area, and location in different hands and hand positions.
[a] ICC $\geq$ 0.7 is desirable and indicates good reliability.

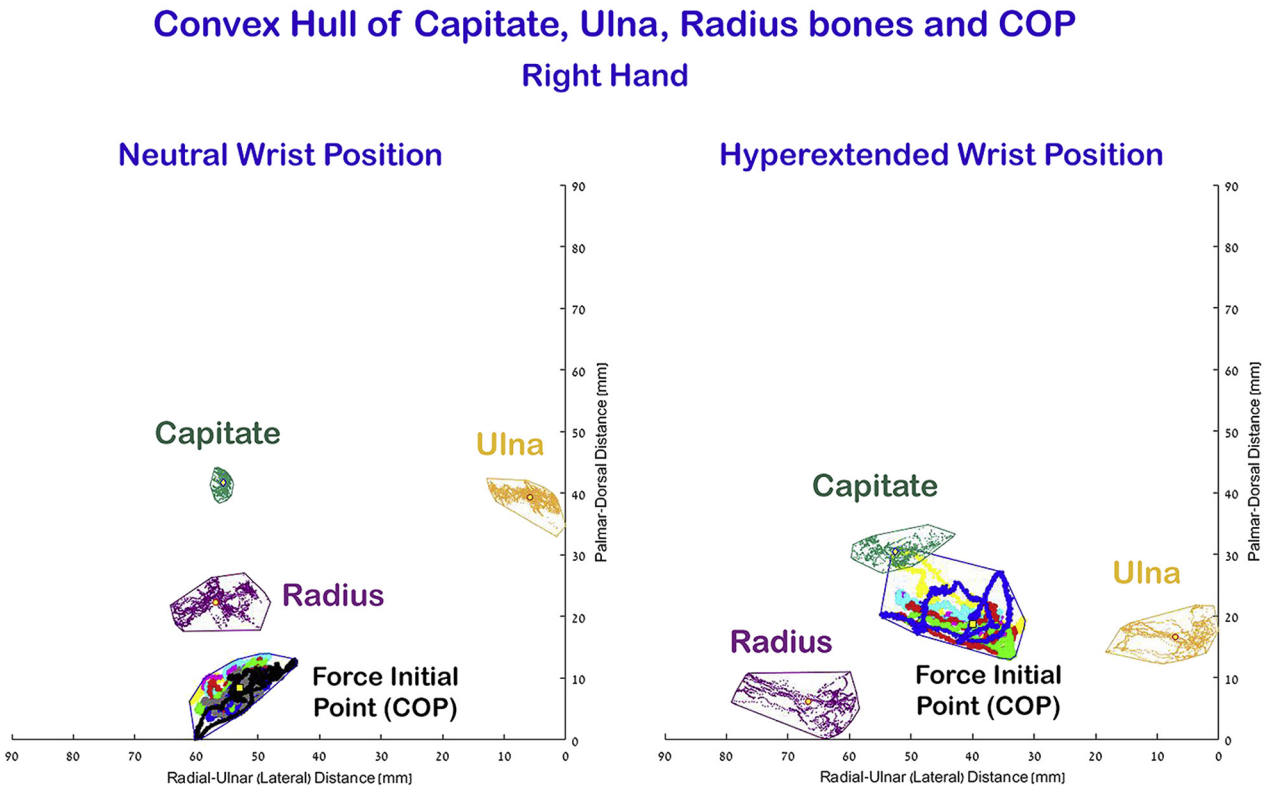


Fig. 2. An example of a projection of measured anatomic markers on the Z = 0 plane and the correspondent convex hull of the location points, hyperextended wrist (HW) vs neutral wrist (NW). The result in the NW position was rotated to be consistent, and all results are shown as dependent on anatomic landmarks only. The distance between the radius and ulna markers did not change throughout the push-up, regardless of the style. The area of the convex hull ground reaction force center of pressure (COP) is significantly smaller for the NW push-up compared with the HW push-up. The forces travel closer to the capitate marker (dorsal) in the hyperextended HW push-up style. The forces also travel closer to the ulna in the HW push-up style.

Discussion

In this study, we compared the dynamic loading, in terms of GRF, of the wrist joint during 2 types of push-ups: push-ups on a HW and push-ups performed on an NW. This study was performed using a noninvasive motion capture system to retrieve kinematic and kinetic parameters during 2 styles of push-ups. A similar technique was used by An et al[16,46] to investigate forces applied to

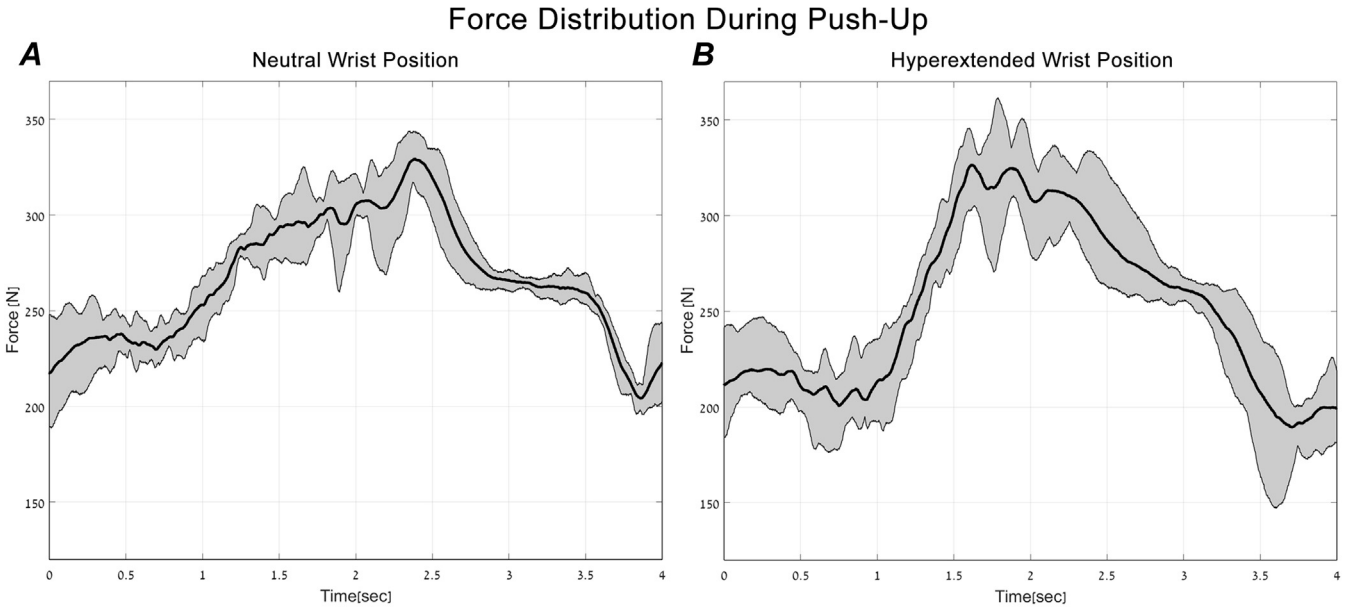


Fig. 3. Example of a typical push-up cycle demonstrating total force changes of a right hand during hyperextended wrist and neutral wrist push-up exercise. (**A**) Neutral wrist position. (**B**) Hyperextended wrist position.

the elbow during push-ups in different hand positions. They concluded that this method enables after dynamic loading with minimal intervention during the exercise under investigation.

We found that, during the push-ups (dynamic wrist loading), an average of 70% of the body weight is loaded on the upper extremity, regardless of style of push-up. This result is compatible with the results reported by Gouvali and Boudolos.[15] They reported that the upper extremity takes an average of 66.4% of the body weight in the HW position. Due to limitations of the study, we were unable to establish the exact travel of the force through the anatomy of the wrist. Because previous studies have demonstrated that forces travel mainly through the radius it is unlikely that forces are shared equally between the radius and ulna at any time during the push-up.[47] However, we did show that forces traveled more ulnarly during push-ups performed in hyperextension than on an NW so it is possible that more force is shared in hyperextension than in neutral.

When observing the geometry of the wrist as indicated by the markers with respect to the GRF location as measured on the force plates, it is evident that there is a difference in the relative location of the GRF and the wrist markers. The location of GRF relative to the capitate, ulna and radius bones effects bone-to-bone force transmission and may cause additional moments in the wrist. In addition, we found that the diversification of the angles between the GRF vector and the normal vector to the force plates during NW push-ups was significantly smaller than those during HW push-ups, meaning that it is likely that there are less sheer forces developed in the wrist during NW push-ups. These sheer forces are the consequence of the projection of the GRF on to the normal tangent direction to the force plates and can affect the wrist bones and soft tissues. More specifically, assuming that all skeletal connective tissues are loaded by GRF, including bones, joints, and ligaments, the non-normal components of GRF may cause increased bending moments on the soft tissues of the joint requiring dynamic compensatory contributions by increasing the cross section of skeletal muscle. The wider distribution of the GRF must be compensated for by the soft tissue and muscles of the upper limbs and trunk muscles.[14] When these muscles are weak, the forces likely transfer to the ligaments and joints.

Furthermore, the convex hull area of the capitate marker trace and the hand markers (positions 11 and 13) were found to be significantly smaller for the NW push-up style compared with the HW style, suggesting more hand motion during the HW push-up style. These results support previous findings related to the significantly smaller angle distribution between the GRF and surface normal during NW push-ups.[22,29,30]

From the geometrical point of view, it was found that the GRFs are more dorsal during HW push-ups compared with the NW configuration (the marker was attached dorsally, and the measured distance was significantly smaller). This observation agrees with those reported in other studies. This observation was not only obtained during dynamic loading but also in the static phases (fully up and fully down) during push-ups.[29,30,32] In addition, we found that the GRF passed close to the radius (through the radius) in the NW push-ups, yet it passed closer to the ulna during HW push-ups. As stated, Rainbow et al demonstrated a 45% increase in contact area between the lunate and radius. This demonstrates the difference in force transmission in the extremely extended wrist and may be a result of more ulnar force during wrist loading.

In light of these findings, it is reasonable that, for the HW push-up style, the forces preferentially pass through ligamentous structures rather than travel directly through bones because they divert more (angle wise) and travel over a wider area. They also travel more ulnarly and dorsally in the wrist, which are areas that are primarily ligamentous (closer to the area of the triangular fibrocartilage complex). On the other hand, when the push-up is performed on the NW, the hand is more stable and the forces seem to pass through areas that are composed of bones and joints. These are more volar and through the radius. It is possible that the forces pass through the capitate and lunate bones and then through the radiolunate and/or radioscaphoid joint into the radius, but this needs to be verified in vivo.

Sports, such as martial arts, emphasize impact on the index and middle metacarpal heads.[48] It has also been established that the index and middle metacarpophalangeal joints are the most stable. It is reasonable, therefore, to assume that most of the force generated during a push-up on an NW passes through this relatively stable anatomic chain into the radius.[49-51] The force plates only provide a calculated COP on the plate, and we are unable to define the exact anatomic trajectory of the forces. We cannot conclude therefore from the study that the index and middle digits are major support points. However, the position of the force in the NW that is closer to the radius is suggestive that most of the forces are in the index and middle digits area. A separate study looking at the transfer of forces through the hand needs to be done to evaluate this assumption.

A limitation of the study is that a direct link between the forces and the bony and ligamentous anatomy could not be established using this methodology. Using this technique, we were unable to establish through which bones and ligaments the forces actually passed through during push-up performance. However, the surface markers did approximate the anatomy and, therefore, give us an idea of the anatomic course of the forces. It is worth mentioning that surface markers are subjected to STAs. The STA errors depend on inertial effects, deformation caused by muscle contraction, skin deformation, and sliding in the area of the joints. In general, they depend on the limb being evaluated and on the task being performed.[52] Most studies found that the rigid displacement of markers with respect to the bone was the dominant component of STA.[44] In our study, the acceleration of the wrist was negligible because the push-ups were performed on a floor, at a low constant rate, and the markers were located close to the bone tissues with minimal movement; hence, a low rate of sliding of soft tissue on the bone. Furthermore, all markers were treated independently and not as a rigid body, and we compared data using the subject as his own control, thus effectively canceling out any effect. We therefore assume that the measurement errors associated with the STA are similar in both push-up methods.

We did not designate specific forearm rotation in this study. The HW push-ups were performed in pronation, whereas the NW push-ups were performed with the forearm in neutral rotation. Although we attempted to correct for forearm rotation by calculating our forces relative to the plane between the markers of the radius and ulna, studies have shown that forearm rotation effects the transfer of forces through the wrist.[53,54] Shaaban et al[54-56] examined transfer of forces through the distal radioulnar joint demonstrating that the position of the distal radioulnar joint affects the transfer of forces through the radius and ulna. It is therefore reasonable to expect an effect on the forces reaching the radiocarpal joint because proximal-distal translation of the radius, which occurs during rotation, influences weight transfer. Specifically, Chen et al[53] as well as Chen and Tang[57] have further demonstrated this. Due to the limitations of this study, we could not evaluate the relative contributions of the wrist/carpus position and the forearm rotations to the forces we measured. However, both these should be taken into account as a component of the push-up performed on an NW and a HW.

Another limitation is that the relationship between our findings in the laboratory and actual clinical consequences is not directly proven but can only be inferred from our knowledge of anatomy

and pathology. In practice, because the forces during the performance of a push-up on an NW are concentrated over the smaller area of the knuckles (metacarpal heads), and because the dorsum of the hand is less suited to loading than the palm, they may be more uncomfortable to perform. Furthermore, hand function, wrist, and forearm rotation may also affect comfort. There are many methods aimed at dissipating these forces, such as bars, handles, and changing the direction of the load.[25,58] An analysis of each of these methods would be helpful in determining the impact of these aids on the transfer of forces and could improve our ability to guide the way push-ups are performed.

This pilot study evaluated the effect of push-ups on the wrist. It is logical to assume that their effect on the other joints in the upper extremity would differ. Because the joints work in tandem—likely in a mechanical chain—we can assume that those forces that are not dissipated around one joint may be transferred to the next. As mentioned in the introduction section, injury to the shoulders and elbows has been described during performance of push-ups.[24] Further study to evaluate the loading of the shoulder and elbow would need to be performed. This is important for rehabilitation of the upper extremity because different exercises would be recommended for different injuries and conditions. For example, if forces are distributed through a larger area in the elbow during push-ups on an NW, patients recovering from an elbow dislocation should be rehabilitated using exercises other than push-ups and certainly not by using push-ups on an NW. The travel of forces through the hand should also be clarified because there are differences that may have clinical consequences. For example, as discussed previously, loading of the metacarpal heads in the NW push-up loads a relatively small area and should be addressed.

In this study, all the subjects were young healthy males in good physical condition. This subject population has the physical ability (muscle mass) to dynamically support the ligamentous structures during performance of push-ups. Further study should not only evaluate the transfer of forces through ligamentous (static stabilizers of joints) but also take into account the dynamic components of push-ups.

This aspect is important for athletes, young military recruits who are starting to build muscle and endurance, as well as for the older population, that is not necessarily in shape but is encouraged by their physicians to exercise. This knowledge is especially important in rehabilitation because the dynamic component is the factor we are able to modify in therapy, and because a better understanding of the patient's abilities and how exercises such as push-ups will affect any particular patient, can improve our ability to tailor the exercise to a particular patient and/or injury.

Conclusion

This study demonstrated that when push-ups are performed in wrist hyperextension with forearms pronated, the forces travel ulnarly and dorsally through tissues such as the triangular fibrocartilage complex, which may not be ideally suited to absorb such forces. In contrast, when push-ups are performed with the wrist and forearm in neutral positions, forces are transferred through the volar wrist and onto the broad articular surface of the distal radius, through the radioscaphoid and radiolunate joints.

Our findings support performance of push-ups on an NW and forearm, especially in the face of wrist ligamentous injury or pain. However, to provide guidelines for strengthening and rehabilitation of the upper extremity in healthy individuals and patients, further study is necessary to delineate differences between types of push-ups as these affect hand, wrist, forearm, elbow, and shoulder. Additional data are needed to determine the precise anatomic locations of force transmission and magnitudes of transmitted loads.

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Quiz: #553

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#1. Force transmission was measured by
   a. surface EMG
   b. video analysis
   c. a system of customized tensiometers
   d. a system of two force plates and a motion sensor

#2. The purpose of the study was to determine
   a. the potential damage of performing push-ups in two different configurations of the wrist
   b. which carpal ligaments are at risk when performing push-ups in two different configurations of the wrist
   c. the force transmission through the wrist during push-ups in two different configurations of the wrist
   d. the stress on the TFCC when performing push-ups in two different configurations of the wrist

#3. The subject population comprised
   a. all normal adults
   b. all patients who had completed their rehab and had successfully returned to work
   c. half teenagers and half senior citizens
   d. 100 adults randomly selected after a Pittsburgh Pirates baseball game

#4. The forces travelled
   a. more volarly and radially in the hyperextended configuration
   b. more dorsally and ulnarly in the hyperextended configuration
   c. more dorsally and ulnarly in the neutral configuration
   d. more distally in both configurations

#5. The data suggest that it is safer to perform push-ups in a neutral wrist arrangement than in the commonly seen hyperextension position
   a. false
   b. true

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